

Ratio-variable predictors: PA size and Mortality

This workbook walks through STATA code evaluate the influence of using PA:AA as a mortality predictor

Objective:

Pulmonary artery diameter on CT scan is a prognostic factor, indicating a higher risk of death when enlarged.

However, body size is a nuisance variable in this evaluation (ie. we don't care about it per se, but want to control for it's relationship with PA size).

On an absolute scale, the pulmonary artery might be large because:

1. The patient has pulmonary hypertension, and pulmonary hypertension is associated with death.
2. They are a larger person, so everything is larger (likely, not associated with death)
3. They are obese, and pulmonary blood flow / thoracic blood volume is larger (may be associated with death through nonspecific mechanisms)

The pulmonary artery to ascending aortic ratio was intended to allow for control of this. (Ng et al. proposed the PA:AA ratio as a CT sign of pulmonary arterial hypertension)

- <https://pubmed.ncbi.nlm.nih.gov/10524808/> - the ratio correlated with mean PA pressure and, unlike absolute PA diameter, showed no association with body surface area (BSA))

By normalizing the PA size to the Ascending aorta size, it accounts for body size (as well as other nuisance variables, like cardiac cycle and image acquisition technique) - A widely cited ATS review stated the aorta serves as an "internal normalization," effectively correcting for BSA, sex, age, cardiac-cycle phase, and technical factors—i.e., normalization beyond body size alone.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4298979/>

However, both the PA diameter and the Ascending Aorta are influenced by age, sex, and comorbidity profiles (which may be distinct). New population analyses questioned the interpretability of PA:AA when aortic caliber varies, arguing that dependence on AA complicates biological meaning and risk signals. This reinforces that the ratio is not a body-size index per se, but a relative vascular-caliber measure influenced by aortic remodeling. <https://pubmed.ncbi.nlm.nih.gov/38637237/>

If AA just functions as a normalization for body size, encoding that information as a single ratio may not discard much information, but if both the PA and the AA encode independent prognostic information, collapsing that down to a single ratio would be potentially misleading.

For example, a normal PA:AA ratio might result from several scenarios, which each potentially having different prognostic implication:

- a small PA and small AA
- a normal PA and normal AA,
- or a large PA and a large AA

We previously reported that both PAd and PA:AA were associated with mortality risk in a cohort of patients who'd receive PA imaging in an emergency room setting.

(<https://onlinelibrary.wiley.com/doi/10.1002/pul2.12225>). In this investigation, we seek re-analyze this data to evaluate whether AA diameter encode additional prognostic information to body size, and whether PA and AA size ought to be modeled separately.

1. Are Cox regression models for hazard of adjusting for age and sex more accurate using PA:AA as a predictor than PA:height, PA:Weight, PA:BMI, PA:BSA, or PA: (Height and Weight)?
2. Are Cox regression models for hazard of death adjusting for PAd, PA:AA substantially less predictive than models that independently model PA and AA

References for the framing:

- Prognostic Factor research recommendations: <https://bmjopen.bmj.com/content/15/8/e095065>
- Ratio variables problematic as predictors:
- Ratio variables problematic as outcomes: <https://doi.org/10.1038/s41366-025-01732-6>

```
In [1]: %pip install stata_setup
import stata_setup
stata_setup.config(r"/Applications/Stata/", "be")
```

Requirement already satisfied: stata_setup in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (0.1.3)

Requirement already satisfied: numpy in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stata_setup) (1.25.2)

Requirement already satisfied: pandas in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stata_setup) (2.0.3)

Requirement already satisfied: ipython in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stata_setup) (9.6.0)

Requirement already satisfied: decorator in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (5.2.1)

Requirement already satisfied: ipython-pygments-lexers in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (1.1.1)

Requirement already satisfied: jedi<=0.16 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (0.19.2)

Requirement already satisfied: matplotlib-inline in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (0.1.7)

Requirement already satisfied: pexpect>4.3 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (4.9.0)

Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (3.0.52)

Requirement already satisfied: pygments>=2.4.0 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (2.19.2)

Requirement already satisfied: stack_data in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (0.6.3)

Requirement already satisfied: traitlets>=5.13.0 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (5.14.3)

Requirement already satisfied: typing_extensions>=4.6 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (4.15.0)

Requirement already satisfied: wcwidth in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython->stata_setup) (0.2.14)

Requirement already satisfied: parso<0.9.0,>=0.8.4 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from jedi<=0.16->ipython->stata_setup) (0.8.5)

Requirement already satisfied: ptyprocess>=0.5 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from pexpect>4.3->ipython->stata_setup) (0.7.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_setup) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_setup) (2025.2)

Requirement already satisfied: tzdata>=2022.1 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_setup) (2025.2)

Requirement already satisfied: six>=1.5 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from python-dateutil>=2.8.2->pandas->stata_setup) (1.17.0)

Requirement already satisfied: executing>=1.2.0 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stack_data->ipython->stata_setup) (2.2.1)

Requirement already satisfied: asttokens>=2.1.0 in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stack_data->ipython->stata_setup) (3.0.0)

Requirement already satisfied: pure_eval in /Users/blocke/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from stack_data->ipython->stata_setup) (0.2.3)

Note: you may need to restart the kernel to use updated packages.



Stata 18.0
BE-Basic Edition

Statistics and Data Science

Copyright 1985–2023 StataCorp LLC

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Notes:

1. Unicode is supported; see help unicode_advice.

In [2]: `%%stata`

```
cd "/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Res
//cd "/Users/reblocke/Research/CTPA-Dose-Response"
```

```
capture mkdir "Results and Figures"
```

```
capture mkdir "Results and Figures/$S_DATE/" //make new folder for figure ou
```

```
capture mkdir "Results and Figures/$S_DATE/Logs/" //new folder for stata log
```

```
local a1=substr(c(current_time),1,2)
```

```
local a2=substr(c(current_time),4,2)
```

```
local a3=substr(c(current_time),7,2)
```

```
//local b = "Body-size Analysis.do" // do file name
```

```
//copy "`b'" "Results and Figures/$S_DATE/Logs/(`a1'`a2'`a3')`b'"
```

```
clear
```

```
set scheme cleanplots //cleanplots white_tableau white_w3d //3 options I use
```

```
use cleaned_noempi
```

```

.
. cd "/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Research
> arch Projects/Pulm Artery Stuff w Scarps/Local Analysis"
/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Research
> rojects/Pulm Artery Stuff w Scarps/Local Analysis

. //cd "/Users/reblocke/Research/CTPA-Dose-Response"
.
. capture mkdir "Results and Figures"

. capture mkdir "Results and Figures/$S_DATE/" //make new folder for figure
out
> put if needed

. capture mkdir "Results and Figures/$S_DATE/Logs/" //new folder for stata l
ogs

. local a1=substr(c(current_time),1,2)

. local a2=substr(c(current_time),4,2)

. local a3=substr(c(current_time),7,2)

. //local b = "Body-size Analysis.do" // do file name
. //copy "`b'" "Results and Figures/$S_DATE/Logs/(`a1'_'a2'_'a3')`b'"
.
. clear

. set scheme cleanplots //cleanplots white_tableau white_w3d //3 options I u
sua
> lly use

.
. use cleaned_noempi
.

```

goal 1: describe the population

Relevant covariates:

PA size (probably want to separate on this by tertile, and a summary paragraph)

- mpad
- age
- male
- height
- weight
- bmi
- death
- time of death

(also comorbidities, for supplement)

Table 1

```
In [3]: %stata
table1_mc, by (mpad_tertile) ///
vars( ///
mpad conts %4.1f \ ///
age conts %4.1f \ ///
male bin %4.1f \ ///
height conts %4.1f \ ///
weight conts %4.1f \ ///
bmi conts %4.1f \ ///
bsa_mosteller conts %4.2f \ ///
ascendingaorta conts %4.1f \ ///
death bin %4.1f \ ///
time_of_death conts %4.1f \ ///
time_of_censoring conts %4.1f \ ///
) ///
percent_n percsign("%") iqrmiddle(",") sdleft(" (±)" sdrightright(")") onecol tot
saving("Results and Figures/$S_DATE/Table 1 Baseline chars by PA tertile.xls
```

```

. table1_mc, by (mpad_tertile) ///
> vars( ///
> mpad conts %4.1f \ ///
> age conts %4.1f \ ///
> male bin %4.1f \ ///
> height conts %4.1f \ ///
> weight conts %4.1f \ ///
> bmi conts %4.1f \ ///
> bsa_mosteller conts %4.2f \ ///
> ascendingaorta conts %4.1f \ ///
> death bin %4.1f \ ///
> time_of_death conts %4.1f \ ///
> time_of_censoring conts %4.1f \ ///
> ) ///
> percent_n percsign("%") iqrmiddle(",") sdleft(" (±)" sdrightright(")") onecol t
ota
> l(before) ///
> saving("Results and Figures/$S_DATE/Table 1 Baseline chars by PA tertile.x
lsx
> ", replace)

```

```

+-----+
---
> -----+
| factor                                N_T  N_1  N_2  N_3  m_T  m_1  m_
2
> m_3 |
|-----+
---
> -----|
| MPAD                                990  336  336  318    0    0
0
> 0 |
|-----+
---
> -----|
| Age (years)                        990  336  336  318    0    0
0
> 0 |
|-----+
---
> -----|
| Male Sex?                          990  336  336  318    0    0
0
> 0 |
|-----+
---
> -----|
| Height (cm)                        829  271  272  286  161   65   6
4
> 32 |
|-----+
---
> -----|
| Weight (kg)                       824  268  271  285  166   68   6
5

```

```

> 33 |
|-----|
-----
> -----|
| BMI (kg/m^2)                818   265   269   284   172   71   6
7
> 34 |
|-----|
-----
> -----|
| BSA (m^2) – Mosteller      816   264   268   284   174   72   6
8
> 34 |
|-----|
-----
> -----|
| Ascending Aorta           990   336   336   318     0     0
0
> 0 |
|-----|
-----
> -----|
| Death                     990   336   336   318     0     0
0
> 0 |
|-----|
-----
> -----|
| Time of Death             263    63    75   125   727   273   26
1
> 193 |
|-----|
-----
> -----|
| Time of longest follow-up alive  727   273   261   193   263    63    7
5
> 125 |
+-----|
-----
> -----+
  N_ ... #records used below,  m_ ... #records not used

+-----|
-----
> -----+
|                                     Total      MPAD Tertile 1
> MPAD Tertile 2      MPAD Tertile 3      p-value |
|-----|
-----
> -----|
|                                     N=990
> N=336              N=318              | N=336
|-----|
-----
> -----|
| MPAD                25.2 (22.5,28.2)      21.5 (20.2,22.5)

```



```

> 25.3 (24.4,26.2)      29.7 (28.3,31.7)      <0.001 |
|-----|
> -----|
| Age (years)          52.0 (37.0,67.0)      45.0 (32.0,59.5)
> 52.0 (36.0,66.0)      59.0 (45.0,73.0)      <0.001 |
|-----|
> -----|
| Male Sex?           37.4% (370)           26.8% (90)
> 36.3% (122)           49.7% (158)           <0.001 |
|-----|
> -----|
| Height (cm)         167.6 (160.0,175.3)    165.1 (157.5,17
2.7)
> 167.6 (160.0,175.3)    170.2 (162.6,177.8)    <0.001 |
|-----|
> -----|
| Weight (kg)         81.9 (68.5,101.4)      76.6 (62.9,89.1)
> 81.2 (70.1,99.2)      91.6 (75.2,113.0)      <0.001 |
|-----|
> -----|
| BMI (kg/m^2)        29.4 (24.6,35.5)      27.8 (23.6,32.6)
> 29.2 (25.1,34.7)      31.9 (25.7,37.7)      <0.001 |
|-----|
> -----|
| BSA (m^2) – Mosteller 1.43 (1.30,1.60)      1.39 (1.29,1.57)
> 1.40 (1.28,1.57)      1.50 (1.34,1.64)      <0.001 |
|-----|
> -----|
| Ascending Aorta     30.1 (27.2,33.3)      27.2 (24.4,30.1)
> 30.1 (27.7,33.0)      32.9 (30.2,35.7)      <0.001 |
|-----|
> -----|
| Death               26.6% (263)           18.8% (63)
> 22.3% (75)            39.3% (125)           <0.001 |
|-----|
> -----|
| Time of Death        3.5 (1.2,6.6)           3.1 (1.0,6.2)
> 4.3 (1.0,8.0)         3.3 (1.3,6.1)           0.45 |
|-----|
> -----|
| Time of longest follow-up alive 10.1 (4.2,11.8) 10.2 (5.1,11.7)
> 9.6 (3.2,11.8)        10.4 (4.2,11.8)        0.67 |
+-----+
> -----+

```

Data are presented as median (IQR) for continuous measures, and % (n) for ca

```
teg
> orical measures.
```

```
file Results and Figures/ 2 Nov 2025/Table 1 Baseline chars by PA tertile.xls
> saved
```

```
.
```

Table 1 supplemental - comorbid conditions

```
In [4]: %stata
table1_mc, by (mpad_tertile) ///
vars( ///
pulmdisease bin %4.1f \ ///
chf bin %4.1f \ ///
diabetes3 cat %4.1f \ ///
hypertension bin %4.1f \ ///
pulmonarycircdisorders bin %4.1f \ ///
peripheralvascdisorders bin %4.1f \ ///
renalfailure bin %4.1f \ ///
anyemergency bin %4.1f \ ///
anyadmission bin %4.1f \ ///
death bin %4.1f \ ///
time_of_death conts %4.1f \ ///
time_of_censoring conts %4.1f \ ///
) ///
percent_n percsign("%") iqrmiddle(",") sdleft(" (±)" sdrightright(")") onecol tot
saving("Results and Figures/$S_DATE/Table s1 comorbs and utilization by PA t
```

```

. table1_mc, by (mpad_tertile) ///
> vars( ///
> pulmdisease bin %4.1f \ ///
> chf bin %4.1f \ ///
> diabetes3 cat %4.1f \ ///
> hypertension bin %4.1f \ ///
> pulmonarycircdisorders bin %4.1f \ ///
> peripheralvascdisorders bin %4.1f \ ///
> renalfailure bin %4.1f \ ///
> anyemergency bin %4.1f \ ///
> anyadmission bin %4.1f \ ///
> death bin %4.1f \ ///
> time_of_death conts %4.1f \ ///
> time_of_censoring conts %4.1f \ ///
> ) ///
> percent_n percsign("%") iqrmiddle(",") sdleft(" (±)" sdrright(")") onecol t
ota
> l(before) ///
> saving("Results and Figures/$S_DATE/Table s1 comorbs and utilization by PA
te
> rtile.xlsx", replace)

```

```

+-----+
---
> -----+
| factor                                N_T   N_1   N_2   N_3   m_T   m_1
> m_2   m_3 |
|-----|
---
> -----|
| Pulmonary Disease                    947   317   320   310   43   19
> 16     8 |
|-----|
---
> -----|
| Congestive Heart Failure              947   317   320   310   43   19
> 16     8 |
|-----|
---
> -----|
| Diabetes                             947   317   320   310   43   19
> 16     8 |
|-----|
---
> -----|
| Hypertension                         947   317   320   310   43   19
> 16     8 |
|-----|
---
> -----|
| Pulm Circ. Disorder                  947   317   320   310   43   19
> 16     8 |
|-----|
---
> -----|
| Periph Vasc. Disease                 947   317   320   310   43   19

```

```

> 16      8 |
> |-----|
> |-----|
> | Kidney Disease                      947   317   320   310   43   19
> 16      8 |
> |-----|
> |-----|
> | Any Emergency Visits in Follow-up?  990   336   336   318     0     0
> 0      0 |
> |-----|
> |-----|
> | Any Hospital Admission in Follow-up? 990   336   336   318     0     0
> 0      0 |
> |-----|
> |-----|
> | Death                              990   336   336   318     0     0
> 0      0 |
> |-----|
> |-----|
> | Time of Death                      263    63    75   125   727   273
> 261   193 |
> |-----|
> |-----|
> | Time of longest follow-up alive      727   273   261   193   263    63
> 75   125 |
> +-----+
> |-----+
> N_ ... #records used below,  m_ ... #records not used
> +-----+
> |-----+
> |                               Total                MPAD Tertile 1
> MPAD Tertile 2  MPAD Tertile 3  p-value |
> |-----|
> |-----|
> |                               N=990                N=336
> N=336           N=318           |
> |-----|
> |-----|
> | Pulmonary Disease                51.0% (483)        47.6% (151)
> 50.9% (163)    54.5% (169)        0.23 |
> |-----|
> |-----|
> | Congestive Heart Failure          18.0% (170)        9.1% (29)
> 13.1% (42)    31.9% (99)         <0.001 |

```

```

-----
> -----|
| Diabetes
> -----|
| No DM <0.001 |
| 75.8% (718) 82.3% (261)
> 78.8% (252) 66.1% (205) |
| Uncomplicated DM 16.3% (154) 12.0% (38)
> 14.1% (45) 22.9% (71) |
| DM with Complication(s) 7.9% (75) 5.7% (18)
> 7.2% (23) 11.0% (34) |
-----
> -----|
| Hypertension 54.7% (518) 42.6% (135)
> 56.2% (180) 65.5% (203) <0.001 |
-----
> -----|
| Pulm Circ. Disorder 15.1% (143) 12.3% (39)
> 14.1% (45) 19.0% (59) 0.051 |
-----
> -----|
| Periph Vasc. Disease 10.9% (103) 7.6% (24)
> 9.1% (29) 16.1% (50) 0.001 |
-----
> -----|
| Kidney Disease 6.4% (61) 3.8% (12)
> 4.7% (15) 11.0% (34) <0.001 |
-----
> -----|
| Any Emergency Visits in Follow-up? 76.9% (761) 80.4% (270)
> 75.0% (252) 75.2% (239) 0.18 |
-----
> -----|
| Any Hospital Admission in Follow-up? 52.7% (522) 47.6% (160)
> 52.4% (176) 58.5% (186) 0.021 |
-----
> -----|
| Death 26.6% (263) 18.8% (63)
> 22.3% (75) 39.3% (125) <0.001 |
-----
> -----|
| Time of Death 3.5 (1.2,6.6) 3.1 (1.0,6.2)
> 4.3 (1.0,8.0) 3.3 (1.3,6.1) 0.45 |
-----
> -----|
| Time of longest follow-up alive 10.1 (4.2,11.8) 10.2 (5.1,11.7)
> 9.6 (3.2,11.8) 10.4 (4.2,11.8) 0.67 |

```

```

+-----+
----
> -----+
Data are presented as median (IQR) for continuous measures, and % (n) for ca
teg
> orical measures.

file Results and Figures/ 2 Nov 2025/Table s1 comorbs and utilization by PA
ter
> tile.xlsx saved

.

```

Section 1: Does the normalization of PA size to AA diameter encode prognostic information beyond body size?

In [5]: `%%stata`
`regress ascendingaorta height weight age i.male`

```

Source |      SS      df      MS      Number of obs      =      8
-----+-----
Model |  8459.25621      4  2114.81405      Prob > F          =      0.00
Residual | 10616.8705     813  13.0588813      R-squared         =      0.44
-----+-----
Adj R-squared =      0.44
Total | 19076.1267     817  23.3489923      Root MSE        =      3.61

```

```

--
ascendinga~a | Coefficient  Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
height |   .0423051   .0144777     2.92   0.004   .0138871   .07072
weight |   .0435943   .0051217     8.51   0.000   .0335409   .05364
age |   .1475674   .0067296    21.93   0.000   .134358   .16077
male |
Male |   1.589054   .3318471     4.79   0.000   .9376756   2.2404
_cons |   11.19544   2.363829     4.74   0.000   6.555514  15.835

```

Interpretation: those predictors (weight, height, age, male - all assumed to be linear) explain 44.07% of the variation in ascending aorta size... the rest is not explained

```
In [6]: %%stata
        pcorr ascendingaorta height weight age i.male
```

(obs=818)

Partial and semipartial correlations of ascendingaorta with

		Partial	Semipartial	Partial	Semipartial	Significan
ce	Variable	corr.	corr.	corr.^2	corr.^2	val
ue						
	-----+-----					
	height	0.1019	0.0765	0.0104	0.0058	0.00
36	weight	0.2860	0.2227	0.0818	0.0496	0.00
00	age	0.6096	0.5737	0.3716	0.3292	0.00
00	0b.male (dropped)					
00	1.male	0.1656	0.1253	0.0274	0.0157	0.00

Interpretation: of the variance no explained by the other predictors, height explains 0.58% of the remaining variance in ascending aorta size, weight explains 4.96% of the variance, age explains 32.9% of the variance, and sex explains 1.5% of the variance.... ie., age is the strongest correaltate of ascending aorta size. (Semipartial correlation^2 is the relevant column)

```
In [7]: %%stata
/* ===== Age+sex-adjusted comparison of PA normalizations (raw ratios) =====
preserve

* Analytic sample (assumes variables already exist & are cleaned)
drop if missing(mpad, ascendingaorta, age, male, height, weight, bmi, bsa, c

* Survival setup
stset lastfollowupyear, failure(death==1)

* Age spline (Harrell 10/50/90) and basis
quietly _pctile age, p(10 50 90)
local k_age `r(r1)' `r(r2)' `r(r3)'
capture drop age_rc*
mkspline2 age_rc = age, cubic knots(`k_age')

* ----- Raw ratio predictors -----
capture drop r_aa r_h r_w r_bmi r_bsa r_hw pa_hat_hw
gen double r_aa = mpad/ascendingaorta
gen double r_h = mpad/height
gen double r_w = mpad/weight
gen double r_bmi = mpad/bmi
```

```

gen double r_bsa = mpad/bsa

* Both-size normalizer: PA divided by raw-units linear prediction from H+W
quietly regress mpad c.height c.weight
predict double pa_hat_hw, xb
replace pa_hat_hw = . if pa_hat_hw<=0
gen double r_hw = mpad/pa_hat_hw

* ----- Fit each model; collect AIC/BIC/C into locals -----
local rvars "r_aa r_h r_w r_bmi r_bsa r_hw"
local names `"' "PA:AA" "PA:Height" "PA:Weight" "PA:BMI" "PA:BSA" "PA:(Ht,Wt

local i = 0
foreach R of local rvars {
    local ++i
    capture drop rc*
    mkspline2 rc = `R', cubic nknots(4)

    * Baseline: age-spline + sex
    quietly stcox c.(rc*) c.(age_rc*) i.male, vce(robust)

    quietly estat ic
    local A`i' = r(S)[1,5]
    local B`i' = r(S)[1,6]

    local C`i' = .
    capture noisily estat concordance
    if (_rc==0) local C`i' = r(C)
}

* ----- Build a small results dataset and print -----
clear
set obs 6
gen str16 model = ""
gen double AIC = .
gen double BIC = .
gen double C = .

local j = 0
foreach lbl of local names {
    local ++j
    replace model = ``\lbl'"" in `j'
    replace AIC = `A`j'' in `j'
    replace BIC = `B`j'' in `j'
    replace C = `C`j'' in `j'
}

format AIC %9.3f BIC %9.3f C %6.4f
list, abbreviate(16) noobs sepby(model)

restore

```



```

. /* ===== Age+sex-adjusted comparison of PA normalizations (raw ratios) ===
==
> */
. preserve

.
. * Analytic sample (assumes variables already exist & are cleaned)
. drop if missing(mpad, ascendingaorta, age, male, height, weight, bmi, bsa,
de
> ath, lastfollowupyear)
(174 observations deleted)

.
. * Survival setup
. stset lastfollowupyear, failure(death==1)

Survival-time data settings

      Failure event: death==1
Observed time interval: (0, lastfollowupyear]
      Exit on or before: failure

-----
      816  total observations
      30  observations end on or before enter()
-----
      786  observations remaining, representing
      243  failures in single-record/single-failure data
6,113.529  total analysis time at risk and under observation
                                     At risk from t =           0
                                     Earliest observed entry t =       0
                                     Last observed exit t =    12.4819

.
. * Age spline (Harrell 10/50/90) and basis
. quietly _pctile age, p(10 50 90)

. local k_age `r(r1)' `r(r2)' `r(r3)''

. capture drop age_rc*

. mkspline2 age_rc = age, cubic knots(`k_age')

.
. * ----- Raw ratio predictors -----
. capture drop r_aa r_h r_w r_bmi r_bsa r_hw pa_hat_hw

. gen double r_aa = mpad/ascendingaorta

. gen double r_h = mpad/height

. gen double r_w = mpad/weight

. gen double r_bmi = mpad/bmi

. gen double r_bsa = mpad/bsa

```

```

.
. * Both-size normalizer: PA divided by raw-units linear prediction from H+W
. quietly regress mpad c.height c.weight

. predict double pa_hat_hw, xb

. replace pa_hat_hw = . if pa_hat_hw<=0
(0 real changes made)

. gen double r_hw = mpad/pa_hat_hw

.
. * ----- Fit each model; collect AIC/BIC/C into locals -----
. local rvars "r_aa r_h r_w r_bmi r_bsa r_hw"

. local names `"'PA:AA" "PA:Height" "PA:Weight" "PA:BMI" "PA:BSA" "PA:(Ht,
Wt)
> " "'

.
. local i = 0

. foreach R of local rvars {
2.     local ++i
3.     capture drop rc*
4.     mkspline2 rc = `R', cubic nknots(4)
5.
.     * Baseline: age-spline + sex
.     quietly stcox c.(rc*) c.(age_rc*) i.male, vce(robust)
6.
.     quietly estat ic
7.     local A`i' = r(S)[1,5]
8.     local B`i' = r(S)[1,6]
9.
.     local C`i' = .
10.    capture noisily estat concordance
11.    if (_rc==0) local C`i' = r(C)
12. }

```

```

Failure _d: death==1
Analysis time _t: lastfollowupyear

```

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529
Number of orderings as expected (E)	=	106482
Number of tied predictions (T)	=	0

Harrell's C = (E + T/2) / P	=	0.7577
Somers' D	=	0.5154

```

Failure _d: death==1
Analysis time _t: lastfollowupyear

```

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529
Number of orderings as expected (E)	=	107339
Number of tied predictions (T)	=	1

Harrell's C = (E + T/2) / P	=	0.7638
Somers' D	=	0.5276

Failure _d: death==1
 Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529
Number of orderings as expected (E)	=	107752
Number of tied predictions (T)	=	0

Harrell's C = (E + T/2) / P	=	0.7668
Somers' D	=	0.5335

Failure _d: death==1
 Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529
Number of orderings as expected (E)	=	108077
Number of tied predictions (T)	=	0

Harrell's C = (E + T/2) / P	=	0.7691
Somers' D	=	0.5381

Failure _d: death==1
 Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529
Number of orderings as expected (E)	=	106752
Number of tied predictions (T)	=	0

Harrell's C = (E + T/2) / P	=	0.7596
Somers' D	=	0.5193

Failure _d: death==1
 Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

Number of subjects (N)	=	786
Number of comparison pairs (P)	=	140529

Number of orderings as expected (E) = 107603
 Number of tied predictions (T) = 0

Harrell's C = (E + T/2) / P = 0.7657
 Somers' D = 0.5314

```
.
. * ----- Build a small results dataset and print -----
. clear

. set obs 6
Number of observations (_N) was 0, now 6.

. gen str16 model = ""
(6 missing values generated)

. gen double AIC = .
(6 missing values generated)

. gen double BIC = .
(6 missing values generated)

. gen double C = .
(6 missing values generated)

.
. local j = 0

. foreach lbl of local names {
  2.   local ++j
  3.   replace model = "`" `lbl' ""' in `j'
  4.   replace AIC   = `A' `j' '' in `j'
  5.   replace BIC   = `B' `j' '' in `j'
  6.   replace C     = `C' `j' '' in `j'
  7. }
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
(1 real change made)
```

```
(1 real change made)
(1 real change made)
(1 real change made)
```

```
.
. format AIC %9.3f BIC %9.3f C %6.4f
. list, abbreviate(16) noobs sepby(model)
```

model	AIC	BIC	C
PA:AA	2842.223	2870.225	.75772261
PA:Height	2833.113	2861.1149	.76382455
PA:Weight	2826.419	2854.4207	.76675989
PA:BMI	2824.854	2852.8557	.76907258
PA:BSA	2843.873	2871.8743	.75964392
PA: (Ht,Wt)	2830.515	2858.5165	.76569961

```
.
. restore
.
```

Interpretation: all the PA diameter normalization strategies work roughly as well as one another in predicting hazard of death after adjusting for age and sex - with the best performing model actually being PA/BMI... though differences are small.

Part 2 : Does modeling PA size and AA size separately, as opposed to a ratio, improve estimation of mortality risk?

Setup - restricted cubic splines allow predictors to take non-linear functional forms (see this intro: <https://www.bmj.com/content/390/bmj-2024-082440>).

We'll compare two approaches:

- model PA:AA ratio flexibly, assessing how various levels of PA:AA predict mortality risk.
- model PA flexibly, and AA flexibly - and use the model to make predictions about mortality risk.
 - we'll assess whether this model is more accurate than the single model.
 - if it is more accurate, and many patients with the same PA:AA ratio have very different predicted mortality risks, that would suggest that collapsing the PA

and AA to a single PA:AA measure would be inadvisable.

How does PA:AA predict risk?

```
In [9]: %stata
preserve
/* ==== PA:AA ratio curve (rcs=4); adj = rcs(age,3) + sex; xblc-safe ==== */

* Survival setup
stset lastfollowupyear, failure(death==1)

* Age spline (3 knots)
capture drop age_rc*
mkspline2 age_rc = age, cubic nknots(3)

* Integerize PA:AA (e.g., 0.77 -> 77) to avoid binary-decimal issues
capture drop paaa100 rc*
gen long paaa100 = round(mpaaa*100)

* 4-knot spline of the predictor
mkspline2 rc = paaa100, cubic nknots(4)

* Cox model: rcs(PA:AA,4) + rcs(age,3) + sex (NO ht/wt)
stcox c.(rc*) c.(age_rc*) i.male, vce(robust)
gen byte esample = e(sample)

* Range (p1-p99, clamped to 0.60-1.30)
quietly summarize mpaaa if esample, detail
local lo = ceil( max(r(p1)*100, 60) )
local hi = floor(min(r(p99)*100, 130) )

* Evaluate only at observed integerized values in-range
levelsof paaa100 if esample & inrange(paaa100, `lo', `hi'), local(atlist)

* Reference 0.77 => 77
xblc rc*, covname(paaa100) at(`atlist') reference(77) eform ///
    generate(pa or lb ub)

* Back to ratio scale
replace pa = pa/100

twoway ///
    (line lb ub pa, sort lc(black black) lp(longdash longdash)) ///
    (line or pa, sort lc(black) lp(solid)), ///
    yscale(log extend) ///
    ylabel(1 2 4, angle(horiz) format(%2.1fc) labsize(large)) ///
    xlabel(0.6(0.1)1.3, labsize(large)) xmtick(0.6(0.1)1.3) ///
    ytitle("Hazard Ratio of Mortality", size(large)) ///
    xtitle("Pulmonary Artery : Ascending Aorta Ratio", size(large)) ///
    title("Adjusted Mortality Risk by PA:AA", size(vlarge)) ///
    note("Model: rcs(PA:AA, 4 on x100) + rcs(age, 3) + sex; robust SEs", size(
    yline(1, lp(shortdash) lc(gs10)) ///
    xline(0.9, lp(shortdash_dot) lc(gs10)) ///
    text(0.85 1.06 "95% CI (black dashes)", size(medlarge)) ///
```

```
text(2.5  0.77 "Reference = 0.77", size(medlarge)) ///  
scheme(cleanplots)  
  
restore
```

```

. preserve

. /* ==== PA:AA ratio curve (rcs-4); adj = rcs(age,3) + sex; xblc-safe ====
*/
.
. * Survival setup
. stset lastfollowupyear, failure(death==1)

Survival-time data settings

          Failure event: death==1
Observed time interval: (0, lastfollowupyear]
          Exit on or before: failure

-----
          990  total observations
           78  observations end on or before enter()
-----
          912  observations remaining, representing
          263  failures in single-record/single-failure data
        6,981.359 total analysis time at risk and under observation
                                     At risk from t =           0
                               Earliest observed entry t =           0
                               Last observed exit t =       12.4819

.
. * Age spline (3 knots)
. capture drop age_rc*

. mkspline2 age_rc = age, cubic nknots(3)

.
. * Integerize PA:AA (e.g., 0.77 -> 77) to avoid binary-decimal issues
. capture drop paaa100 rc*

. gen long paaa100 = round(mpaaa*100)

.
. * 4-knot spline of the predictor
. mkspline2 rc = paaa100, cubic nknots(4)

.
. * Cox model: rcs(PA:AA,4) + rcs(age,3) + sex    (NO ht/wt)
. stcox c.(rc*) c.(age_rc*) i.male, vce(robust)

          Failure _d: death==1
          Analysis time _t: lastfollowupyear

Iteration 0:  Log pseudolikelihood = -1693.3646
Iteration 1:  Log pseudolikelihood = -1606.9315
Iteration 2:  Log pseudolikelihood = -1567.1941
Iteration 3:  Log pseudolikelihood = -1567.1317
Iteration 4:  Log pseudolikelihood = -1567.1317
Refining estimates:
Iteration 0:  Log pseudolikelihood = -1567.1317

```


Cox regression with Breslow method for ties

```

No. of subjects =      912                Number of obs =      9
12
No. of failures =      263
Time at risk    = 6,981.3588

                                Wald chi2(6)  = 232.
01
Log pseudolikelihood = -1567.1317         Prob > chi2   = 0.00
00

```

```

--

```

	_t	Haz. ratio	Robust std. err.	z	P> z	[95% conf. interval]	
	rc1	1.007172	.0156569	0.46	0.646	.976948	1.0383
31	rc2	1.015181	.0555174	0.28	0.783	.9119971	1.1300
38	rc3	.9930323	.1872283	-0.04	0.970	.6862401	1.436
98	age_rc1	1.060324	.0128577	4.83	0.000	1.035421	1.0858
27	age_rc2	.9970443	.011934	-0.25	0.805	.9739264	1.0207
11							
	male						
47	Male	1.542152	.1923577	3.47	0.001	1.207686	1.9692

```

--

```

```

. gen byte esample = e(sample)

.
. * Range (p1-p99, clamped to 0.60-1.30)
. quietly summarize mpaaa if esample, detail

. local lo = ceil( max(r(p1)*100, 60) )

. local hi = floor(min(r(p99)*100, 130) )

.
. * Evaluate only at observed integerized values in-range
. levelsof paaa100 if esample & inrange(paaa100, `lo', `hi'), local(atlist)
60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 8
5 8
> 6 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 1
08
> 109 110 111 113 114 115 116 117 118 119

.
. * Reference 0.77 => 77

```

```
. xblc rc*, covname(paaa100) at(`atlist') reference(77) eform ///
> generate(pa or lb ub)
```

paaa100	exp(xb)	(95% CI)
60	0.88	(0.56–1.36)
61	0.88	(0.58–1.33)
62	0.89	(0.60–1.30)
63	0.89	(0.63–1.27)
64	0.90	(0.65–1.24)
65	0.91	(0.68–1.22)
66	0.91	(0.70–1.19)
67	0.92	(0.73–1.16)
68	0.93	(0.76–1.14)
69	0.93	(0.78–1.11)
70	0.94	(0.81–1.09)
71	0.95	(0.84–1.07)
72	0.96	(0.87–1.05)
73	0.96	(0.90–1.04)
74	0.97	(0.92–1.02)
75	0.98	(0.95–1.01)
76	0.99	(0.97–1.01)
77	1.00	(1.00–1.00)
78	1.01	(1.00–1.03)
79	1.02	(0.99–1.05)
80	1.03	(0.99–1.08)
81	1.05	(0.99–1.11)
82	1.06	(0.98–1.15)
83	1.08	(0.97–1.20)
84	1.09	(0.96–1.24)
85	1.11	(0.95–1.30)
86	1.13	(0.94–1.36)
87	1.15	(0.94–1.42)
88	1.17	(0.93–1.48)
89	1.20	(0.93–1.55)
90	1.22	(0.93–1.61)
91	1.25	(0.93–1.67)
92	1.28	(0.94–1.74)
93	1.31	(0.95–1.80)
94	1.34	(0.97–1.85)
95	1.38	(0.99–1.91)
96	1.41	(1.01–1.97)
97	1.45	(1.04–2.02)
98	1.49	(1.07–2.08)
99	1.53	(1.10–2.14)
100	1.58	(1.13–2.20)
101	1.62	(1.17–2.26)
102	1.67	(1.20–2.33)
103	1.72	(1.24–2.40)
104	1.78	(1.27–2.48)
105	1.83	(1.31–2.56)
106	1.89	(1.34–2.66)
107	1.94	(1.37–2.76)
108	2.00	(1.40–2.87)
109	2.06	(1.42–3.00)
110	2.13	(1.45–3.13)
111	2.19	(1.47–3.27)

113	2.33	(1.51–3.59)
114	2.40	(1.53–3.77)
115	2.48	(1.55–3.96)
116	2.55	(1.57–4.16)
117	2.63	(1.58–4.37)
118	2.71	(1.60–4.60)
119	2.79	(1.61–4.85)

```

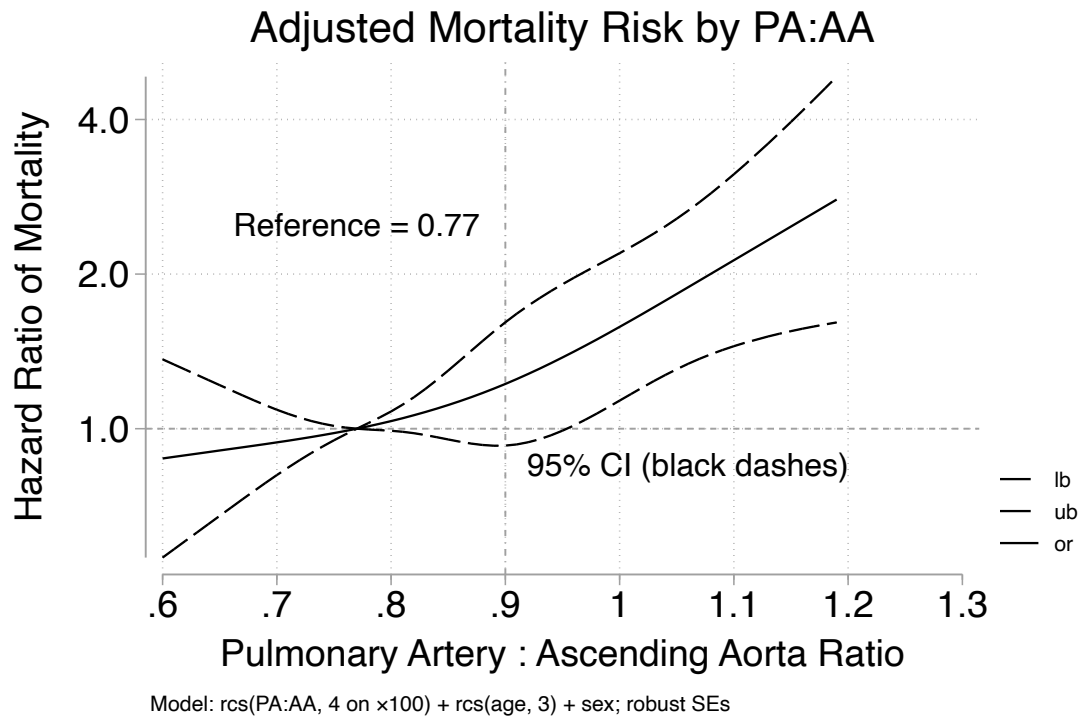
.
. * Back to ratio scale
. replace pa = pa/100
(59 real changes made)

.
. twoway ///
> (line lb ub pa, sort lc(black black) lp(longdash longdash)) ///
> (line or pa, sort lc(black) lp(solid)), ///
> yscale(log extend) ///
> ylabel(1 2 4, angle(horiz) format(%2.1fc) labsize(large)) ///
> xlabel(0.6(0.1)1.3, labsize(large)) xmtick(0.6(0.1)1.3) ///
> ylabel("Hazard Ratio of Mortality", size(large)) ///
> xlabel("Pulmonary Artery : Ascending Aorta Ratio", size(large)) ///
> title("Adjusted Mortality Risk by PA:AA", size(vlarge)) ///
> note("Model: rcs(PA:AA, 4 on x100) + rcs(age, 3) + sex; robust SEs", siz
e(s
> mall)) ///
> yline(1, lp(shortdash) lc(gs10)) ///
> xline(0.9, lp(shortdash_dot) lc(gs10)) ///
> text(0.85 1.06 "95% CI (black dashes)", size(medlarge)) ///
> text(2.5 0.77 "Reference = 0.77", size(medlarge)) ///
> scheme(cleanplots)

.
. restore

.

```



Heatmap of the predictions of the flexibly modeled splines

```
In [10]: %stata
preserve
/* ===== Heatmap: observed cells only | rcs(age,3); sex averaged | NO ht/wt

* 1) Setup + knots
stset lastfollowupyear, failure(death==1)

capture drop mpad_rounded aa_rounded
gen mpad_rounded = round(mpad, 0.1)
gen aa_rounded   = round(ascendingaorta, 0.1)

quietly _pctile mpad_rounded if !missing(mpad_rounded), p(10 50 90)
local k_mpad `r(r1)' `r(r2)' `r(r3)''
quietly summarize mpad_rounded, meanonly
local mp_min = r(min)
local mp_max = r(max)

quietly _pctile aa_rounded if !missing(aa_rounded), p(10 50 90)
local k_aa `r(r1)' `r(r2)' `r(r3)''
quietly summarize aa_rounded, meanonly
local aa_min = r(min)
local aa_max = r(max)

quietly _pctile age if !missing(age), p(10 50 90)
local k_age `r(r1)' `r(r2)' `r(r3)''
quietly summarize age, meanonly
local age_min = r(min)
local age_max = r(max)
```

```

* 2) Fit 2D model: rcs(PA,3)*rcs(AA,3) + rcs(age,3) + sex
capture drop mpad_rc* aa_rc* age_rc*
mkspline2 mpad_rc = mpad_rounded, cubic knots(`k_mpad')
mkspline2 aa_rc   = aa_rounded,   cubic knots(`k_aa')
mkspline2 age_rc  = age,           cubic knots(`k_age')

stcox c.(mpad_rc*)##c.(aa_rc*) c.(age_rc*) i.male, vce(robust)
estimates store M1
estat concordance
gen byte esample = e(sample)

* Holds for prediction (means on the estimation sample)
quietly summarize age if esample, meanonly
scalar AGEbar = r(mean)
quietly summarize male if esample, meanonly
scalar PMALE  = r(mean)

* Means of PA and AA (for reference point)
quietly summarize mpad if esample, meanonly
scalar MPABar = r(mean)
quietly summarize ascendingaorta if esample, meanonly
scalar AABar  = r(mean)

* 3) Observed-cell grid only
tempfile EST
save `EST', replace
use `EST', clear
keep if esample
gen mpad_cell = round(mpad, 1)
gen aa_cell   = round(ascendingaorta, 1)
contract mpad_cell aa_cell
rename mpad_cell mpad
rename aa_cell   ascendingaorta
rename _freq     Ncell

* dynamic axes from observed cells
quietly summarize mpad, meanonly
local xlo = floor(r(min)) // PA for Y axis
local xhi = ceil(r(max))
quietly summarize ascendingaorta, meanonly
local ylo = floor(r(min)) // AA for X axis
local yhi = ceil(r(max))

* duplicate per sex; age at mean
expand 2
bysort mpad ascendingaorta: gen male = _n-1
gen age = AGEbar

* add sentinels (ensure >2 age values for spline basis)
local N0 = _N
set obs `=_N + 2'
replace mpad           = 25          in `=' `N0' + 1'
replace ascendingaorta = 30          in `=' `N0' + 1'
replace age            = `age_min' in `=' `N0' + 1'
replace male           = 0           in `=' `N0' + 1'

```

```

replace mpad          = 25          in `=' `N0' + 2'
replace ascendingaorta = 30          in `=' `N0' + 2'
replace age           = `age_max' in `=' `N0' + 2'
replace male          = 0           in `=' `N0' + 2'
gen byte sentinel = (_n > `N0')

* 4) Predict on the grid; sex-average; normalize to (MPAbar, AAbars)
gen mpad_rounded = round(mpad, 0.1)
gen aa_rounded   = round(ascendingaorta, 0.1)
mkspline2 mpad_rc = mpad_rounded, cubic knots(`k_mpad')
mkspline2 aa_rc   = aa_rounded,   cubic knots(`k_aa')
mkspline2 age_rc  = age,          cubic knots(`k_age')

estimates restore M1
predict xb, xb
gen HR = exp(xb)

drop if sentinel==1

gen w = cond(male==1, PMALE, 1-PMALE)
bysort mpad ascendingaorta: egen HRw = total(HR*w)
bysort mpad ascendingaorta: egen wsum = total(w)
gen HR_avg = HRw/wsum

gen double d2 = (mpad - MPAbar)^2 + (ascendingaorta - AAbars)^2
bysort mpad ascendingaorta: keep if _n==1
sort d2
scalar HRREF          = HR_avg[1]
local ref_m_used = mpad[1]
local ref_a_used = ascendingaorta[1]
gen double normalized_HR = HR_avg/HRREF

* 5) Bin + labeled legend
capture drop hr_bin
local cuts "0.55 0.67 0.83 0.91 1 1.10 1.25 1.50 2 3 4 6 9 12 16.5"
egen hr_bin = cut(normalized_HR), at(`cuts') icodes
label define hr_lab 1 "<=0.67" 2 "0.67-0.83" 3 "0.83-0.91" 4 "0.91-1.00" ///
                    5 "1.00-1.10" 6 "1.10-1.25" 7 "1.25-1.50" 8 "1.50-2.00"
                    9 "2.00-3.00" 10 "3.00-4.00" 11 "4.00-6.00" 12 "6.00-9.00"
                    13 "9.00-12.00" 14 ">=12.00", replace
label values hr_bin hr_lab

levelsof hr_bin if hr_bin>0 & !missing(hr_bin), local(usedbins)
local ramp_labels
foreach k of local usedbins {
    local lab : label (hr_bin) `k'
    local ramp_labels `ramp_labels' `k' "`lab'"
}
local nbins : word count `usedbins'

* ratio guide lines + reference marker (X=AA, Y=PA)
local overlays
foreach r in 0.7 0.8 0.9 1.0 1.1 {
    local overlays `overlays' || ///
        function y = `r'*x, range(`ylo' `yhi') ///
        lcolor(black%35) lpattern(shortdash) lwidth(thin) legend(off)

```

```

}
local overlays `overlays' || ///
    scatter mpad ascendingaorta if mpad==`ref_m_used' & ascendingaorta==`ref
    msymbol(plus) mcolor(black) msize(medlarge) legend(off)

* 6) Heatmap
heatmap hr_bin mpad ascendingaorta, discrete ///
    color(RdYlBu, reverse n(`nbins')) ///
    ramp(right subtitle("HR (vs ref)") label(`ramp_labels')) ///
    title("Predicted Hazard Ratio of Death (vs mean)") ///
    subtitle("AA and PA + interaction (age at mean; sex averaged)", size(med
    xtitle("Ascending Aorta Diameter (mm)") ///
    ytitle("Main PA Diameter (mm)") ///
    xlabel(`ylo'(5)`yhi') ylabel(`xlo'(5)`xhi') ///
    scheme(white_w3d) ///
    addplot(`overlays')

* sanity checks
di "Ref cell used: PA=" `ref_m_used' " AA=" `ref_a_used'
list normalized_HR if mpad==`ref_m_used' & ascendingaorta==`ref_a_used'
di "Mean PA:AA ratio = " %5.3f (MPAbar/AAbar)

restore

```

```

. preserve

. /* ===== Heatmap: observed cells only | rcs(age,3); sex averaged | NO ht/w
t =
> ===== */
.
. * 1) Setup + knots
. stset lastfollowupyear, failure(death==1)

Survival-time data settings

      Failure event: death==1
Observed time interval: (0, lastfollowupyear]
      Exit on or before: failure

-----
      990  total observations
      78  observations end on or before enter()
-----
      912  observations remaining, representing
      263  failures in single-record/single-failure data
6,981.359  total analysis time at risk and under observation
                                     At risk from t =           0
                                     Earliest observed entry t =       0
                                     Last observed exit t =    12.4819

.
. capture drop mpad_rounded aa_rounded

. gen mpad_rounded = round(mpad, 0.1)

. gen aa_rounded    = round(ascendingaorta, 0.1)

.
. quietly _pctile mpad_rounded if !missing(mpad_rounded), p(10 50 90)

. local k_mpad `r(r1)' `r(r2)' `r(r3)''

. quietly summarize mpad_rounded, meanonly

. local mp_min = r(min)

. local mp_max = r(max)

.
. quietly _pctile aa_rounded if !missing(aa_rounded), p(10 50 90)

. local k_aa `r(r1)' `r(r2)' `r(r3)''

. quietly summarize aa_rounded, meanonly

. local aa_min = r(min)

. local aa_max = r(max)

.

```



```

. quietly _pctile age if !missing(age), p(10 50 90)

. local k_age `r(r1)' `r(r2)' `r(r3)''

. quietly summarize age, meanonly

. local age_min = r(min)

. local age_max = r(max)

.

. * 2) Fit 2D model: rcs(PA,3)*rcs(AA,3) + rcs(age,3) + sex
. capture drop mpad_rc* aa_rc* age_rc*

. mkspline2 mpad_rc = mpad_rounded, cubic knots(`k_mpad')

. mkspline2 aa_rc   = aa_rounded,   cubic knots(`k_aa')

. mkspline2 age_rc  = age,           cubic knots(`k_age')

.

. stcox c.(mpad_rc*)##c.(aa_rc*) c.(age_rc*) i.male, vce(robust)

      Failure _d: death==1
      Analysis time _t: lastfollowupyear

```

```

Iteration 0: Log pseudolikelihood = -1693.3646
Iteration 1: Log pseudolikelihood = -1603.1223
Iteration 2: Log pseudolikelihood = -1562.9013
Iteration 3: Log pseudolikelihood = -1562.7593
Iteration 4: Log pseudolikelihood = -1562.7593
Refining estimates:
Iteration 0: Log pseudolikelihood = -1562.7593

```

Cox regression with Breslow method for ties

```

No. of subjects =          912                      Number of obs =      9
12
No. of failures =           263
Time at risk    = 6,981.3588

Wald chi2(11) = 243.

17
Log pseudolikelihood = -1562.7593                    Prob > chi2    = 0.00
00

```

```

-----
--
      _t |
      | Haz. ratio   Robust
      |             std. err.      z    P>|z|      [95% conf. interval]
-----+-----
01  mpad_rc1 |    1.945967    1.214407    1.07   0.286    .5727054    6.6121
93  mpad_rc2 |    1.450567    1.28598    0.42   0.675    .2552214    8.2443

```

44	aa_rc1		1.769719	.8828832	1.14	0.253	.6656623	4.7049
76	aa_rc2		.4615428	.2787557	-1.28	0.200	.1412915	1.5076
	c.mpad_rc1#							
24	c.aa_rc1		.9753996	.0216278	-1.12	0.261	.9339177	1.0187
	c.mpad_rc1#							
61	c.aa_rc2		1.033659	.026566	1.29	0.198	.9828803	1.0870
	c.mpad_rc2#							
56	c.aa_rc1		.9923225	.0305085	-0.25	0.802	.9342928	1.0539
	c.mpad_rc2#							
86	c.aa_rc2		.9913503	.0285352	-0.30	0.763	.9369706	1.0488
	age_rc1		1.0485	.0144945	3.43	0.001	1.020472	1.0772
97	age_rc2		1.005424	.0131572	0.41	0.679	.9799641	1.0315
45	male							
08	Male		1.428554	.1911061	2.67	0.008	1.099073	1.8568

. estimates store M1

. estat concordance

Failure _d: death==1
Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

Number of subjects (N) = 912
Number of comparison pairs (P) = 174623
Number of orderings as expected (E) = 132591
Number of tied predictions (T) = 0

Harrell's C = (E + T/2) / P = 0.7593
Somers' D = 0.5186

. gen byte esample = e(sample)

.
. * Holds for prediction (means on the estimation sample)
. quietly summarize age if esample, meanonly

. scalar AGEbar = r(mean)

```

. quietly summarize male if esample, meanonly

. scalar PMALE = r(mean)

.
. * Means of PA and AA (for reference point)
. quietly summarize mpad if esample, meanonly

. scalar MPABar = r(mean)

. quietly summarize ascendingaorta if esample, meanonly

. scalar AABar = r(mean)

.
. * 3) Observed-cell grid only
. tempfile EST

. save `EST', replace
(file /var/folders/vf/n84t7b8171lf64smq4ddvw1m0000gn/T//St35529.00000a not
 found)
file /var/folders/vf/n84t7b8171lf64smq4ddvw1m0000gn/T//St35529.00000a saved
 as .dta format

. use `EST', clear

. keep if esample
(78 observations deleted)

. gen mpad_cell = round(mpad, 1)

. gen aa_cell = round(ascendingaorta, 1)

. contract mpad_cell aa_cell

. rename mpad_cell mpad

. rename aa_cell ascendingaorta

. rename _freq Ncell

.
. * dynamic axes from observed cells
. quietly summarize mpad, meanonly

. local xlo = floor(r(min)) // PA for Y axis

. local xhi = ceil(r(max))

. quietly summarize ascendingaorta, meanonly

. local ylo = floor(r(min)) // AA for X axis

. local yhi = ceil(r(max))

```

```

.
. * duplicate per sex; age at mean
. expand 2
(293 observations created)

. bysort mpad ascendingaorta: gen male = _n-1

. gen age = AGEbar

.
. * add sentinels (ensure >2 age values for spline basis)
. local N0 = _N

. set obs `=_N + 2'
Number of observations (_N) was 586, now 588.

. replace mpad          = 25          in `=' `N0' + 1'
(1 real change made)

. replace ascendingaorta = 30          in `=' `N0' + 1'
(1 real change made)

. replace age           = `age_min' in `=' `N0' + 1'
(1 real change made)

. replace male          = 0           in `=' `N0' + 1'
(1 real change made)

.
. replace mpad          = 25          in `=' `N0' + 2'
(1 real change made)

. replace ascendingaorta = 30          in `=' `N0' + 2'
(1 real change made)

. replace age           = `age_max' in `=' `N0' + 2'
(1 real change made)

. replace male          = 0           in `=' `N0' + 2'
(1 real change made)

. gen byte sentinel = (_n > `N0')

.
. * 4) Predict on the grid; sex-average; normalize to (MPAbar, AAbars)
. gen mpad_rounded = round(mpad, 0.1)

. gen aa_rounded   = round(ascendingaorta, 0.1)

. mkspline2 mpad_rc = mpad_rounded, cubic knots(`k_mpad')

. mkspline2 aa_rc   = aa_rounded,   cubic knots(`k_aa')

. mkspline2 age_rc  = age,           cubic knots(`k_age')

.

```

```

. estimates restore M1
(results M1 are active now)

. predict xb, xb

. gen HR = exp(xb)

.
. drop if sentinel==1
(2 observations deleted)

.
. gen w = cond(male==1, PMALE, 1-PMALE)

. bysort mpad ascendingaorta: egen HRw = total(HR*w)

. bysort mpad ascendingaorta: egen wsum = total(w)

. gen HR_avg = HRw/wsum

.
. gen double d2 = (mpad - MPABar)^2 + (ascendingaorta - AABar)^2

. bysort mpad ascendingaorta: keep if _n==1
(293 observations deleted)

. sort d2

. scalar HRREF      = HR_avg[1]

. local  ref_m_used = mpad[1]

. local  ref_a_used = ascendingaorta[1]

. gen double normalized_HR = HR_avg/HRREF

.
. * 5) Bin + labeled legend
. capture drop hr_bin

. local cuts "0.55 0.67 0.83 0.91 1 1.10 1.25 1.50 2 3 4 6 9 12 16.5"

. egen hr_bin = cut(normalized_HR), at(`cuts') icodes
(7 missing values generated)

. label define hr_lab 1 "<=0.67" 2 "0.67-0.83" 3 "0.83-0.91" 4 "0.91-1.00"
///
>                    5 "1.00-1.10" 6 "1.10-1.25" 7 "1.25-1.50" 8 "1.50-2.0
0" /
> //
>                    9 "2.00-3.00" 10 "3.00-4.00" 11 "4.00-6.00" 12 "6.00-
9.00
> " ///
>                    13 "9.00-12.00" 14 ">=12.00", replace

. label values hr_bin hr_lab

```

```

.
. levelsof hr_bin if hr_bin>0 & !missing(hr_bin), local(usedbins)
1 2 3 4 5 6 7 8 9 10 11 12

. local ramp_labels

. foreach k of local usedbins {
2.     local lab : label (hr_bin) `k'
3.     local ramp_labels `ramp_labels' `k' ``lab'
4. }

. local nbins : word count `usedbins'

.
. * ratio guide lines + reference marker (X=AA, Y=PA)
. local overlays

. foreach r in 0.7 0.8 0.9 1.0 1.1 {
2.     local overlays `overlays' || ///
>         function y = `r'*x, range(`ylo' `yhi') ///
>         lcolor(black%35) lpattern(shortdash) lwidth(thin) legend(off)
3. }

. local overlays `overlays' || ///
>     scatter mpad ascendingaorta if mpad==`ref_m_used' & ascendingaorta==`r
ef_
> a_used', ///
>     msymbol(plus) mcolor(black) msize(medlarge) legend(off)

.
. * 6) Heatmap
. heatmap hr_bin mpad ascendingaorta, discrete ///
>     color(RdYlBu, reverse n(`nbins')) ///
>     ramp(right subtitle("HR (vs ref)") label(`ramp_labels')) ///
>     title("Predicted Hazard Ratio of Death (vs mean)") ///
>     subtitle("AA and PA + interaction (age at mean; sex averaged)", size(m
eds
> mall)) ///
>     xtitle("Ascending Aorta Diameter (mm)") ///
>     ytitle("Main PA Diameter (mm)") ///
>     xlabel(`ylo'(5)`yhi') ylabel(`xlo'(5)`xhi') ///
>     scheme(white_w3d) ///
>     addplot(`overlays')
(note:  named style i not found in class symbol, default attributes used)

.
. * sanity checks
. di "Ref cell used:  PA=" `ref_m_used' "  AA=" `ref_a_used'
Ref cell used:  PA=26  AA=30

. list normalized_HR if mpad==`ref_m_used' & ascendingaorta==`ref_a_used'

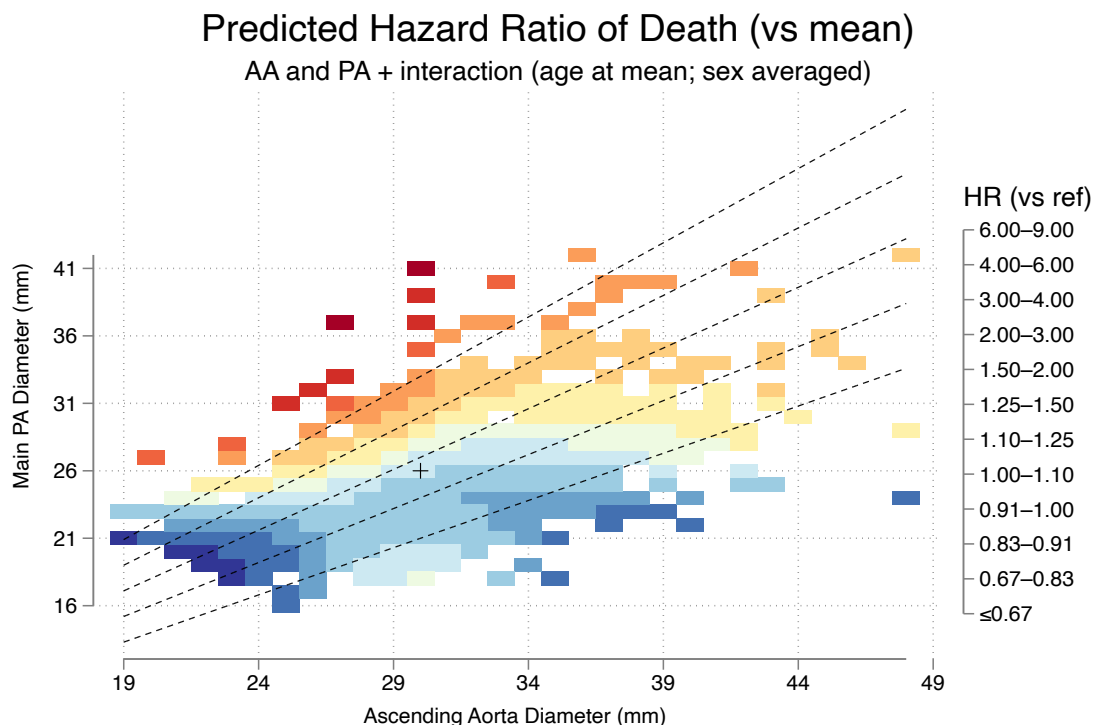
+-----+
| normal~R |
|-----|

```

```
1. |           1 |
   +-----+
```

```
. di "Mean PA:AA ratio = " %5.3f (MPAbar/AAbar)
Mean PA:AA ratio = 0.844
```

```
.
. restore
```



Interpretation: the predicted risk for patients on the same PA:AA line can vary quite a lot, whether it's "small PA and small AA", "normal normal", or "big big"

Does the dual spline model predict much more accurately than the single ratio model?

```
In [11]: %stata
preserve
/* ==== Model comparison: 1D ratio vs 2D PA×AA (adj = rcs(age,3) + sex) ====
stset lastfollowupyear, failure(death==1)

* Age spline
capture drop age_rc*
mkspline2 age_rc = age, cubic nknots(3)

* --- M_Ratio: rcs(PA:AA,4) on ×100 ---
capture drop paaa100 rcR*
gen long paaa100 = round(mpaaa*100)
mkspline2 rcR = paaa100, cubic nknots(4)
stcox c.(age_rc*) i.male c.(rcR*), vce(robust)
```

```

tempname S aR bR cR
quietly estat ic
matrix `S' = r(S)
scalar `aR' = `S'[1,5]
scalar `bR' = `S'[1,6]
capture noisily estat concordance
scalar `cR' = .
if (_rc==0) scalar `cR' = r(C)
capture drop rcR*

* --- M_2D: rcs(PA,3) × rcs(AA,3) + rcs(age,3) + sex
quietly _pctile mpad, p(10 50 90)
local k_mpad `r(r1)' `r(r2)' `r(r3)''
quietly _pctile ascendingaorta, p(10 50 90)
local k_aa `r(r1)' `r(r2)' `r(r3)''

capture drop mpad_rc* aa_rc*
mkspline2 mpad_rc = mpad, cubic knots(`k_mpad')
mkspline2 aa_rc = ascendingaorta, cubic knots(`k_aa')

stcox c.(mpad_rc*)##c.(aa_rc*) c.(age_rc*) i.male, vce(robust)

tempname S2 a2 b2 c2
quietly estat ic
matrix `S2' = r(S)
scalar `a2' = `S2'[1,5]
scalar `b2' = `S2'[1,6]
capture noisily estat concordance
scalar `c2' = .
if (_rc==0) scalar `c2' = r(C)

* Report
matrix M = ( scalar(`aR') , scalar(`bR') , scalar(`cR') \ ///
             scalar(`a2') , scalar(`b2') , scalar(`c2') )
matrix rownames M = M_Ratio M_2D
matrix colnames M = AIC BIC C
di as txt "Model comparison: Ratio (1D) vs 2D surface [adj: rcs(age,3)+sex]"
matlist M, format(%9.4f)

* Clean
capture drop mpad_rc* aa_rc* age_rc* paaa100
restore

```



```
. preserve
```

```
. /* ==== Model comparison: 1D ratio vs 2D PA×AA (adj = rcs(age,3) + sex) ==
```

```
> */
```

```
. stset lastfollowupyear, failure(death==1)
```

Survival-time data settings

Failure event: death==1

Observed time interval: (0, lastfollowupyear]

Exit on or before: failure

```
-----
          990  total observations
```

```
          78  observations end on or before enter()
```

```
-----
          912  observations remaining, representing
```

```
          263  failures in single-record/single-failure data
```

```
        6,981.359  total analysis time at risk and under observation
```

```
                        At risk from t =          0
```

```
                Earliest observed entry t =          0
```

```
                Last observed exit t =    12.4819
```

```
.
```

```
. * Age spline
```

```
. capture drop age_rc*
```

```
. mkspline2 age_rc = age, cubic nknots(3)
```

```
.
```

```
. * --- M_Ratio: rcs(PA:AA,4) on ×100 ---
```

```
. capture drop paaa100 rcR*
```

```
. gen long paaa100 = round(mpaaa*100)
```

```
. mkspline2 rcR = paaa100, cubic nknots(4)
```

```
. stcox c.(age_rc*) i.male c.(rcR*), vce(robust)
```

Failure _d: death==1

Analysis time _t: lastfollowupyear

```
Iteration 0:  Log pseudolikelihood = -1693.3646
```

```
Iteration 1:  Log pseudolikelihood = -1606.9315
```

```
Iteration 2:  Log pseudolikelihood = -1567.1941
```

```
Iteration 3:  Log pseudolikelihood = -1567.1317
```

```
Iteration 4:  Log pseudolikelihood = -1567.1317
```

Refining estimates:

```
Iteration 0:  Log pseudolikelihood = -1567.1317
```

Cox regression with Breslow method for ties

No. of subjects = 912

Number of obs = 9

12

No. of failures = 263

Time at risk = 6,981.3588

Wald chi2(6) = 232.

01

Log pseudolikelihood = -1567.1317

Prob > chi2 = 0.00

00

		Haz. ratio	Robust std. err.	z	P> z	[95% conf. interval]	
	age_rc1	1.060324	.0128577	4.83	0.000	1.035421	1.0858
	age_rc2	.9970443	.011934	-0.25	0.805	.9739264	1.0207
	male						
	Male	1.542152	.1923577	3.47	0.001	1.207686	1.9692
	rcR1	1.007172	.0156569	0.46	0.646	.976948	1.0383
	rcR2	1.015181	.0555174	0.28	0.783	.9119971	1.1300
	rcR3	.9930323	.1872283	-0.04	0.970	.6862401	1.436

```

.
. tempname S aR bR cR

. quietly estat ic

. matrix `S' = r(S)

. scalar `aR' = `S'[1,5]

. scalar `bR' = `S'[1,6]

. capture noisily estat concordance

      Failure _d: death==1
      Analysis time _t: lastfollowupyear

```

Harrell's C concordance statistic

```

Number of subjects (N)      =      912
Number of comparison pairs (P) =    174623
Number of orderings as expected (E) =  132027
Number of tied predictions (T) =       19

```

```

      Harrell's C = (E + T/2) / P =    0.7561
      Somers' D =    0.5122

```

```

. scalar `cR' = .

. if (_rc==0) scalar `cR' = r(C)

. capture drop rcR*

.
. * --- M_2D: rcs(PA,3) × rcs(AA,3) + rcs(age,3) + sex
. quietly _pctile mpad, p(10 50 90)

. local k_mpad `r(r1)' `r(r2)' `r(r3)''

. quietly _pctile ascendingaorta, p(10 50 90)

. local k_aa `r(r1)' `r(r2)' `r(r3)''

.
. capture drop mpad_rc* aa_rc*

. mkspline2 mpad_rc = mpad,          cubic knots(`k_mpad')

. mkspline2 aa_rc   = ascendingaorta, cubic knots(`k_aa')

.
. stcox c.(mpad_rc*)##c.(aa_rc*) c.(age_rc*) i.male, vce(robust)

```

Failure _d: death==1
 Analysis time _t: lastfollowupyear

```

Iteration 0: Log pseudolikelihood = -1693.3646
Iteration 1: Log pseudolikelihood = -1603.1223
Iteration 2: Log pseudolikelihood = -1562.9013
Iteration 3: Log pseudolikelihood = -1562.7593
Iteration 4: Log pseudolikelihood = -1562.7593
Refining estimates:
Iteration 0: Log pseudolikelihood = -1562.7593

```

Cox regression with Breslow method for ties

```

No. of subjects =      912                      Number of obs =      9
12
No. of failures =      263
Time at risk    = 6,981.3588

Wald chi2(11) = 243.

17
Log pseudolikelihood = -1562.7593              Prob > chi2    = 0.00
00

```

```

-----
--
          |
          | Robust
          | Haz. ratio  std. err.      z    P>|z|      [95% conf. interval]
          |-----+-----
          |
          | mpad_rc1 |      1.945967      1.214407      1.07    0.286      .5727054      6.6121

```

01							
	mpad_rc2	1.450567	1.28598	0.42	0.675	.2552214	8.2443
93							
	aa_rc1	1.769719	.8828832	1.14	0.253	.6656623	4.7049
44							
	aa_rc2	.4615428	.2787557	-1.28	0.200	.1412915	1.5076
76							
	c.mpad_rc1#						
	c.aa_rc1	.9753996	.0216278	-1.12	0.261	.9339177	1.0187
24							
	c.mpad_rc1#						
	c.aa_rc2	1.033659	.026566	1.29	0.198	.9828803	1.0870
61							
	c.mpad_rc2#						
	c.aa_rc1	.9923225	.0305085	-0.25	0.802	.9342928	1.0539
56							
	c.mpad_rc2#						
	c.aa_rc2	.9913503	.0285352	-0.30	0.763	.9369706	1.0488
86							
	age_rc1	1.0485	.0144945	3.43	0.001	1.020472	1.0772
97							
	age_rc2	1.005424	.0131572	0.41	0.679	.9799641	1.0315
45							
	male						
	Male	1.428554	.1911061	2.67	0.008	1.099073	1.8568
08							

```

.
. tempname S2 a2 b2 c2

. quietly estat ic

. matrix `S2' = r(S)

. scalar `a2' = `S2'[1,5]

. scalar `b2' = `S2'[1,6]

. capture noisily estat concordance

      Failure _d: death==1
      Analysis time _t: lastfollowupyear

```

Harrell's C concordance statistic

```

Number of subjects (N)          =      912
Number of comparison pairs (P)   =    174623
Number of orderings as expected (E) =    132591

```

Number of tied predictions (T) = 0

Harrell's C = (E + T/2) / P = 0.7593

Somers' D = 0.5186

```
. scalar `c2' = .

. if (_rc==0) scalar `c2' = r(C)

.
. * Report
. matrix M = ( scalar(`aR') , scalar(`bR') , scalar(`cR') \ ///
>              scalar(`a2') , scalar(`b2') , scalar(`c2') )

. matrix rownames M = M_Ratio M_2D

. matrix colnames M = AIC BIC C

. di as txt "Model comparison: Ratio (1D) vs 2D surface [adj: rcs(age,3)+se
x]"
Model comparison: Ratio (1D) vs 2D surface [adj: rcs(age,3)+sex]

. matlist M, format(%9.4f)
```

	AIC	BIC	C
M_Ratio	3146.2634	3175.1572	0.7561
M_2D	3147.5185	3200.4906	0.7593

```
.
. * Clean
. capture drop mpad_rc* aa_rc* age_rc* paaa100

. restore

.
```

Interpretation: the dual spline model is a little bit more accurate, but not enough to justify the use of all the additional degrees of freedom (model flexibility)

Overall conclusion:

1. AA mostly depends on age in this sample, with smaller independent contributions from sex, height, and weight... but a little more than half the variation is not explained by (linear increases in) those factors
2. In age, sex adjusted models that use PA size to predict hazard of death, normalizing by metrics of body-size works roughly as well as normalizing by AA... suggesting that AA is not encoding substantial additional prognostic information in this group.
3. If PA size and AA size are both modeled flexibly, the predicted hazard of death can vary substantially among patients with the same PA:AA values - depending on which type of PA and AA values (small small, med med, large large). However, the more

complicated model is not all that much more predictive than the simple one - suggesting that in practice this distinction doesn't have much consequence.

Therefore, as most of the prognostic information is encoded in the PA diameter itself - parsimonious models should focus on PA_diameter +/- one of the body size adjustments.